Balancing Consistency and Generalization via Cross-Sample Memory Bank for Remote Sensing Image Change Detection

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***Abstract*—Change detection (CD) is a fundamental task in remote sensing image analysis, aiming to identify changes between images acquired at different times. For CD, it is critical to learn the consistent features of unchanged regions across multitemporal images. However, forcing consistent features also impacts the representation of diverse scenarios, resulting in decayed generalization capability. In this article, we simultaneously consider the unchanged consistency and cross-scene generalization and propose a constraint and memory network (C-MNet). First, a dual-branch backbone integrating SAM2 and VMamba is constructed to model both spatial and cross-temporal dependencies simultaneously for better capturing the consistency. Then a mask-guided dual-temporal reconstruction module is designed which enforces cross-temporal reconstruction consistency on unchanged areas and enables the network to explicitly learn temporal invariance. However, consistency-driven reconstruction tends to suppress feature diversity in different scenarios, biasing the model toward dominant land-cover patterns. To overcome this limitation, we further design a cross-sample Memory Bank mechanism that stores stable unchanged features and serves as a historical reference during training. C-MNet effectively balances consistency constraints and feature diversity, leading to improved recognition capability of changed regions and cross-scene generalization. The code of this work can be found on https://github.com/4043-lab/Change-detection**

***Index Terms*—Change detection (CD), cross-temporal consistency, memory bank, reconstruction constraint, remote sensing image, SAM2, VMamba.**

I. Introduction

CHANGE detection (CD) in remote sensing aims to identify land cover changes between images acquired at different times. It supports a wide range of applications such as urban expansion monitoring [1], [2], [3], [4], disaster assessment [5], land-use updating [6], [7], [8], and ecological supervision. Compared with natural images, remote sensing CD faces more complex imaging conditions and land-cover structures: illumination, seasonal variations, sensor discrepancies, and registration errors across time can substantially alter pixel appearances, making it difficult to distinguish appearance changes from real changes. Therefore, high-quality CD requires not only accurate localization of changed areas but also robust cross-scene generalization under diverse scene distributions.

Deep learning has significantly advanced remote sensing CD in recent years. Existing methods can be broadly categorized into two types. Difference-based methods extract features from bi-temporal images separately [9], [10], [11] and explicitly construct change representations via feature differencing [12], [13], [14], absolute differences [15], [16], or concatenation [17], [18], followed by a classification or segmentation head to predict change masks [19], [20]. While intuitive, differencing may amplify both nuisance variations and real changes, resulting in false alarms and missed detections. Joint encoding and decoding methods perform joint modeling of bi-temporal images, e.g., shared or coupled encoders, cross-temporal interaction attention, and fused decoders, to learn cross-temporal dependencies and correspondences within the network [15], [21], [22]. This paradigm has become a dominant direction in modern CD.

CD relies on separability between changed regions and unchanged regions. High-quality prediction typically requires strong unchanged consistency, namely learning temporally invariant representations for unchanged areas to avoid confusing illumination or seasonal variations with real land-cover changes [23], [24]. However, when consistency is continuously reinforced, the model may collapse unchanged representations into overly similar embeddings, smoothing out genuine variations within unchanged regions, weakening the ability to capture scene diversity, and eventually degrading cross-scene generalization.

To improve robustness, many studies focus on stronger consistency constraints, such as reconstruction objectives [25], contrastive objectives [26], or cross-temporal regularization to enforce stable representations for unchanged areas [27]. While these strategies can reduce pseudo-changes caused by imaging variations, their hidden cost is often underestimated: strong constraints may oversmooth features in unchanged regions and weaken the representation of scene diversity. Moreover, unchanged regions do not always imply homogeneity, as the scenes shown in Fig. 1. For example, unchanged forests near urban boundaries and those in primitive forests exhibit markedly different structural and textural statistics. If a model

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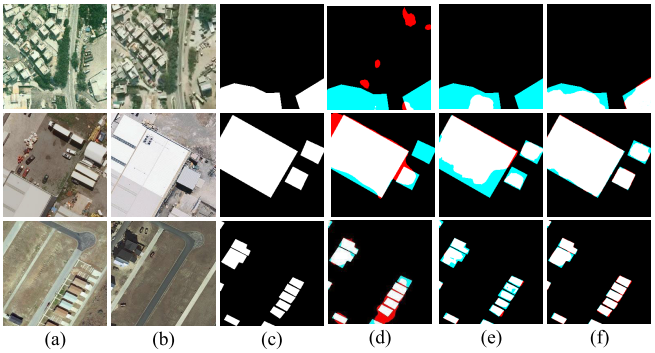


Fig. 1. Qualitative ablation of the proposed modules. White, black, red, and cyan indicate TP, TN, FP, and FN, respectively. (a) T1, (b) T2, (c) GT, (d) baseline (w/o Recon and Memory), (e) +Recon (showing suppressed pseudo-changes but slight oversmoothing), and (f) ours (combining Recon and Memory to achieve both unchanged consistency and changed completeness).

primarily pursues consistency, it can become biased toward dominant land-cover patterns in the training set, leading to poor generalization on unseen scenes. An approach to improve generalization is to use memory [28] or prototype mechanisms [29], which are widely used in classification and segmentation to preserve stable representations, but they are seldom designed specifically for CD, especially for modeling unchanged-region representations. As shown in Fig. 1, by designing and applying a memory bank in CD, many missed detections can be avoided across different scenes.

Focusing on the above discussions, we propose a novel CD network to balance the consistency and feature diversity, thereby improving cross-scene generalization. The main contributions of this work can be summarized as follows.

1) We propose the constraint and memory network (C-MNet), with a dual-branch backbone integrating SAM2 [30] and VMamba [31] to jointly model spatial and cross-temporal dependencies, thereby strengthening consistency-oriented representation learning for unchanged regions.
2) We design a mask-guided dual-temporal reconstruction module that enforces cross-temporal reconstruction consistency on unchanged areas, enabling the network to explicitly learn temporal invariance and reduce pseudo-changes.
3) We design a cross-sample memory bank that stores stable sample-level representations across scenes and provides historical references for memory-guided regularization.
4) Extensive comparisons with state-of-the-art methods and ablation studies on the Wuhan University Dataset (WHU-CD), Sun Yat-sen University dataset (SYSU-CD), and LEVIR-CD datasets demonstrate consistent improvements across multiple metrics, confirming the effectiveness, competitiveness, and generalization capability of the proposed approach.

The remainder of this article is organized as follows. Section II reviews related work. Section III formulates the problem and presents the overall C-MNet framework, including its key components. Section IV describes the experimental settings, comparative results, and ablation studies. Section V concludes this article and discusses future work.

II. Related Works

A. Change Detection

CD is a fundamental task in remote sensing image analysis, aiming to identify land cover changes between images acquired at different times [5], [32]. Traditional methods of CD can be broadly divided into two categories: difference-based methods and joint encoding methods. Each approach has its strengths and weaknesses, especially in modeling unchanged regions, which are crucial for distinguishing real changes from nuisances such as illumination or seasonal variations.

CNN-Based Methods: Traditional CNN-based methods, such as FC-Siam-diff and FC-Siam-conc (Siamese architecture), utilize two identical network branches for encoding the two temporal images separately [9], [10], [33]. The difference or concatenation of features between the two branches is then used to predict change maps. These methods struggle to properly handle unchanged regions, often leading to false positives (FPs) at boundaries or regions where changes are subtle.

Siamese Networks: SRCDNet [12], [34], [35] is a Siamese-based network designed for CD, specifically in images of varying resolutions. It uses a super-resolution module to first enhance low-resolution images before feeding them into a Siamese network with attention mechanisms for feature extraction. However, it still faces the issue of oversmoothing when enforcing consistency in unchanged regions.

Transformer-Based Methods: With the success of Transformers, self-attention mechanisms have been introduced into CD. BIT [36] models temporal dependencies using compact visual tokens, while PA-Former [21] and SwinSUNet [37] further incorporate hierarchical attention. Recent studies also explore training Transformers from scratch [38] and integrating vision-language knowledge [22]. Although Transformer-based methods exhibit strong modeling capacity, they may still struggle to capture fine-grained variations in unchanged regions.

Despite significant progress, most existing methods focus primarily on modeling changed regions and lack explicit mechanisms to preserve the diversity of unchanged regions under complex imaging conditions [39], [40].

B. Reconstruction in Remote Sensing

Reconstruction-based learning has been widely explored in remote sensing and computer vision tasks, including image restoration, domain adaptation, and representation learning [41], [42]. In CD, reconstruction is commonly employed as an auxiliary supervision to enforce temporal consistency.

Most reconstruction-based CD frameworks follow an encoder–decoder paradigm. Bi-temporal images are first mapped into latent feature spaces using shared or partially shared encoders, and reconstruction decoders are then utilized

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to recover the original inputs [25], [27]. By minimizing reconstruction errors, the encoder is encouraged to suppress noise caused by illumination variations and sensor discrepancies.

Some methods perform independent reconstruction for each temporal image [26], while others adopt cross-temporal reconstruction strategies [25], where features from one temporal image are used to reconstruct the other. This design explicitly promotes temporal invariance.

To prevent reconstruction from interfering with real changes, selective constraints are often adopted. For example, reconstruction losses may be applied to shallow features or masked unchanged regions [20], [39]. Different from existing strategies, we propose a mask-guided dual-temporal reconstruction combined with cross-temporal consistency specifically on shallow features. This design explicitly enforces temporal invariance on unchanged regions while robustly preserving fine-grained appearance details. However, although such reconstruction-based methods provide dense self-supervision and effectively enhance temporal consistency, relying solely on them inherently favors feature similarity and stability, which may limit their ability to preserve diverse scene characteristics.

To overcome this dilemma specifically for CD, our method uniquely introduces a cross-sample memory mechanism to retrieve diverse historical representations, successfully balancing temporal consistency with cross-scene discriminability.

C. Memory and Prototype Learning

Memory-based learning has been extensively studied in deep learning to store and reuse representative features across samples and training iterations. In remote sensing, memory mechanisms have been introduced to improve robustness under limited annotations and domain shifts [29].

In typical memory-based frameworks, feature embeddings extracted from training samples are stored in memory banks and retrieved through similarity matching [28]. These memory entries serve as historical references and enable long-range contextual modeling beyond individual mini-batches.

Recent studies further integrate neural memory into Transformer architectures for CD. ChangeTitans [43] and BiSAM-CD [28] exploit temporal memory to suppress pseudo-changes and enhance temporal consistency. Memory-supported Transformers such as MS-Former [29] demonstrate improved performance under weak supervision.

Prototype learning represents a related paradigm, where feature distributions are summarized using a small set of representative vectors. This strategy has been shown to stabilize feature learning and reduce intraclass variance in remote sensing tasks [29].

However, memory-based mechanisms for CD remain relatively under-explored. Existing designs mainly focus on modeling changed regions and do not explicitly tailor memory structures for representing diverse unchanged regions. Moreover, without temporal stabilization, memory banks may absorb time-specific noise, degrading generalization performance.

Therefore, effective memory mechanisms for CD should jointly preserve temporal stability and cross-scene diversity.

III. Method

A. Overview

Given a bi-temporal remote sensing image pair $I^{t_1}, I^{t_2} \in \mathbb{R}^{3\times H\times W}$, CD aims to predict a pixel-wise binary change map $\hat{Y} \in \mathbb{R}^{2\times H\times W}$ indicating changed and unchanged regions. In practical scenarios, a reliable detector must meet two competing requirements. It should learn temporally invariant representations on unchanged regions to suppress pseudo-changes caused by illumination, seasonal, and atmospheric variations. Meanwhile, it should retain sufficient feature diversity across scenes, because unchanged areas may exhibit large appearance differences under different land-cover contexts, and overly enforcing invariance can collapse the representation toward dominant training patterns, degrading cross-scene generalization.

To address this, we propose a C-MNet that balances temporal consistency and cross-scene diversity within a unified pipeline. C-MNet contains a dual-branch backbone, a mask-guided dual-temporal reconstruction constraint, and a cross-sample hierarchical memory bank. As shown in Fig. 2, the overall data flow is summarized as follows.

Spatial features are extracted by a SAM2 branch, which provides strong multiscale priors and robust geometric cues. To preserve the generality of these priors, the SAM2 trunk is frozen, and lightweight adapters are introduced to adapt the model to remote sensing CD with minimal trainable parameters. In parallel, a VMamba branch encodes bi-temporal images and produces multiscale features that efficiently capture long-range dependencies. Cross-temporal interactions are injected stage-wise before selective scanning, strengthening temporal modeling without modifying the scan kernel.

Based on VMamba features from two temporal inputs, C-MNet computes stage-wise temporal difference representations that highlight change-sensitive cues while preserving contextual information. These temporal difference features are then fused with SAM2 spatial features at corresponding scales by cross-modal attention and gated fusion, resulting in fused multiscale representations $\{\mathbf{F}_i\}$. A change decoder aggregates $\{\mathbf{F}_i\}$ and outputs the final change map $\hat{\mathbf{Y}}$. Deep supervision heads attached to $\{\mathbf{F}_i\}$ provide auxiliary predictions to stabilize optimization and improve gradient propagation.

Beyond the detection stream, C-MNet introduces two auxiliary mechanisms. A dual-temporal reconstruction branch reconstructs $\hat{\mathbf{I}}^{t_1}$ and $\hat{\mathbf{I}}^{t_2}$ from shallow VMamba patch features, and applies mask-guided constraints only over unchanged pixels to explicitly enforce temporal invariance. A cross-sample hierarchical memory bank stores compact sample-level memory features extracted from selected VMamba stages. During training, a cross-sample hierarchical memory bank stores compact sample-level representations extracted from selected VMamba stages. For each current sample, relevant historical representations are retrieved to construct a memory-guided regularization term, which encourages the model to preserve diverse cross-sample feature patterns while maintaining temporal stability. The memory bank is used only during training and introduces no additional memory retrieval or update operations during inference.

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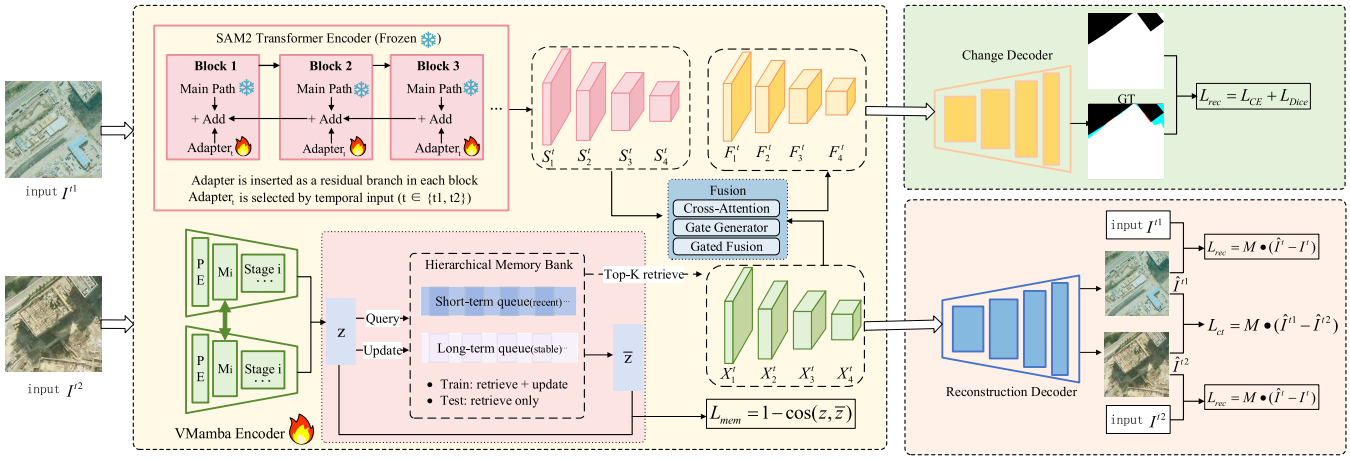


Fig. 2. Overview of the proposed C-MNet for bi-temporal remote sensing CD. C-MNet adopts a dual-branch backbone consisting of a frozen SAM2 spatial branch and a VMamba temporal branch with stage-wise cross-temporal interaction. Multiscale temporal difference features from VMamba are fused with SAM2 spatial features to produce change-aware representations, which are aggregated by the change decoder to predict the final change map. A mask-guided dual-temporal reconstruction branch enforces temporal invariance on unchanged regions during training. Meanwhile, a cross-sample hierarchical memory bank provides historical feature references to balance consistency and feature diversity and improve cross-scene generalization. The fire icon indicates trainable parameters, while the blue snowflake icon indicates frozen parameters.

B. SAM2–VMamba Dual-Branch Backbone

The dual-branch backbone is designed to combine complementary strengths. The SAM2 branch provides strong multiscale spatial priors and robust structural cues. The VMamba branch provides efficient long-range modeling and is enhanced to incorporate cross-temporal interactions before selective scanning. Together, they produce spatially grounded and temporally aware representations for change decoding.

1) SAM2 Spatial Branch With Lightweight Dual Adapters: We adopt the trunk of SAM2 image encoder to extract multiscale spatial features for each temporal image

$$\{\mathbf{S}_i^{t_1}\}_{i=1}^4, \{\mathbf{S}_i^{t_2}\}_{i=1}^4, \quad \mathbf{S}_i^t \in \mathbb{R}^{C_i^{\text{sam}} \times H_i \times W_i}. \tag{1}$$

To preserve the generality of SAM2 priors, the trunk parameters are frozen. Instead of full fine-tuning, we insert lightweight adapters into each transformer block. In implementation, each block contains two small adapter pathways that can be switched according to the temporal input, enabling efficient domain adaptation for both t_1 and t_2 while keeping the backbone stable. This design makes SAM2 act as a strong spatial anchor with minimal additional training cost.

2) Cross-Temporal VMamba Branch: VMamba encodes each temporal image with shared parameters and outputs a patch-embedding feature and multistage encoder features

$$\mathbf{X}_0^t \in \mathbb{R}^{C_0^m \times H_0 \times W_0}, \quad \mathbf{X}_i^t \in \mathbb{R}^{C_i^m \times H_i \times W_i}, \quad i > 0. \tag{2}$$

A key limitation of standard single-image scanning is that each temporal stream is processed independently. To strengthen temporal coupling without modifying the scan kernel, we inject cross-temporal interactions stage-wise before each VMamba encoding stage. Let $\tilde{\mathbf{U}}_i^{t_1}$ and $\tilde{\mathbf{U}}_i^{t_2}$ denote the NHWC features at stage i input. A lightweight temporal mixer $\mathcal{M}_i$ performs gated fusion with residual refinement

$$\bar{\mathbf{U}}_i^{t_1} = \mathcal{M}_i\left(\tilde{\mathbf{U}}_i^{t_1}, \tilde{\mathbf{U}}_i^{t_2}\right), \quad \bar{\mathbf{U}}_i^{t_2} = \mathcal{M}_i\left(\tilde{\mathbf{U}}_i^{t_2}, \tilde{\mathbf{U}}_i^{t_1}\right). \tag{3}$$

The mixed features are then fed into the original VMamba stage

$$\mathbf{U}_{i+1}^{t_1} = \text{VSSLayer}_i\left(\bar{\mathbf{U}}_i^{t_1}\right), \quad \mathbf{U}_{i+1}^{t_2} = \text{VSSLayer}_i\left(\bar{\mathbf{U}}_i^{t_2}\right). \tag{4}$$

This mechanism ensures that each stage receives cross-temporal context prior to scanning, yielding temporally aware representations while remaining compatible with pretrained VMamba weights. We place the cross-temporal interaction before selective scanning so that information from the two temporal inputs can participate in the subsequent state propagation process; otherwise, postscanning interaction would only fuse independently scanned features and could not influence the internal long-range modeling of VMamba.

3) Temporal Difference Modeling: Based on VMamba multiscale features from both temporal inputs, C-MNet computes stage-wise temporal difference features $\{\mathbf{D}_i\}$ that emphasize change cues. The temporal difference module combines complementary difference signals, including absolute difference cues, concatenation-based convolutional cues, and attention-guided difference cues. These signals are adaptively fused by learnable weights normalized by Softmax, producing a robust difference representation for each stage. Importantly, the difference features are projected to a unified channel dimension, providing consistent interfaces for subsequent fusion and decoding.

4) Multiscale Fusion Between SAM2 and VMamba: At each stage, SAM2 spatial features and VMamba temporal difference features are aligned to the same spatial resolution and mapped to a unified channel dimension. We adopt a cross-modal attention module where SAM2 features act as the query and temporal difference features act as key and value, enabling spatial priors to selectively attend to change-sensitive patterns. A gated fusion unit further controls the contribution of the fused signal and the spatial anchor feature, producing fused stage features $\{\mathbf{F}_i\}$ used for change decoding and deep supervision.

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5) Change Decoding and Deep Supervision: The fused multiscale features $\{\mathbf{F}_i\}$ are fed into a dedicated change decoder that aggregates multiscale information through progressive upsampling and feature enhancement. The decoder outputs the final change logits $\hat{\mathbf{Y}}$. In addition, lightweight stage heads generate auxiliary logits at each stage and are upsampled to the original resolution to compute deep supervision losses. This design improves optimization stability and encourages each scale to learn discriminative change-aware representations.

C. Mask-Guided Dual-Temporal Reconstruction

Remote sensing CD is frequently affected by pseudo-changes introduced by illumination shifts, seasonal variations, atmospheric conditions, and sensor differences. These factors often dominate the appearance difference in unchanged areas and lead to false alarms. Although the main change supervision provides dense pixel-level guidance, it does not explicitly enforce temporal invariance in unchanged regions. To explicitly enhance unchanged consistency, we introduce a dual-temporal reconstruction constraint as an auxiliary branch. The key principle is that unchanged regions provide dense self-supervision: if a pixel is unchanged, its content should be recoverable from temporally invariant features, and reconstructions across two times should be consistent.

1) Reconstruction Branch: C-MNet attaches a lightweight reconstruction decoder to shallow VMamba patch features. Let $\mathbf{P}^{t_1}$ and $\mathbf{P}^{t_2}$ denote patch-level features extracted from the VMamba patch embedding stage. A shared decoder $\mathcal{D}_{rec}$ reconstructs both temporal images

$$\hat{\mathbf{I}}^{t_1} = \mathcal{D}_{rec}\left(\mathbf{P}^{t_1}\right), \quad \hat{\mathbf{I}}^{t_2} = \mathcal{D}_{rec}\left(\mathbf{P}^{t_2}\right) \tag{5}$$

where $\hat{\mathbf{I}}^{t_1}, \hat{\mathbf{I}}^{t_2} \in \mathbb{R}^{3\times H\times W}$. Using shallow features is intentional because they preserve more appearance details, making reconstruction a reliable signal to regularize early representations while leaving high-level change reasoning to the detection head.

2) Mask-Guided Reconstruction on Unchanged Areas: Given the ground-truth change label $\mathbf{Y} \in \{0,1\}^{H\times W}$, the unchanged mask is defined as $\mathbf{M} = \mathbb{I}(\mathbf{Y} = 0) \in \{0,1\}^{H\times W}$. We enforce reconstruction only on unchanged pixels to avoid suppressing true changes. The mask-guided reconstruction loss is

$$\mathcal{L}_{rec} = \frac{1}{\mu(\mathbf{M})}\left(\left\|\mathbf{M}\odot\left(\hat{\mathbf{I}}^{t_1} - \mathbf{I}^{t_1}\right)\right\|_1 + \left\|\mathbf{M}\odot\left(\hat{\mathbf{I}}^{t_2} - \mathbf{I}^{t_2}\right)\right\|_1\right) \tag{6}$$

where $\odot$ denotes element-wise multiplication. $\mu(\mathbf{M}) = 1/HW\sum_{h,w}\mathbf{M}_{h,w} + \varepsilon$ is the mean value of the mask used for normalization, ensuring stable loss magnitude even when unchanged pixels occupy a small fraction.

a) Cross-temporal reconstruction consistency: Relying solely on single-image reconstruction may allow the decoder to fit appearance noise independently at each time, which does not necessarily enforce temporal invariance. Therefore, we further impose a cross-temporal reconstruction consistency constraint on unchanged areas

$$\mathcal{L}_{ct} = \frac{1}{\mu(\mathbf{M})}\left\|\mathbf{M}\odot\left(\hat{\mathbf{I}}^{t_1} - \hat{\mathbf{I}}^{t_2}\right)\right\|_1. \tag{7}$$

This term explicitly aligns the reconstructed contents across time and discourages the network from encoding time-specific nuisance factors in unchanged representations.

The proposed constraint branch provides two complementary effects. First, it supplies dense supervision in unchanged regions, which strengthens temporal invariance and suppresses pseudo-changes. Second, by masking out changed pixels, it avoids oversmoothing real changes, thus keeping the discriminability of change features. It should be noted that changed pixels are not ignored in the overall optimization. Although they are excluded from the reconstruction constraint, they are still explicitly supervised by the main CD loss (L_{main}), which is computed over the entire change map, and further optimized by the deep supervision loss (L_{ds}) at multiple feature levels. Therefore, the reconstruction branch mainly regularizes unchanged-region consistency, while the detection branch ensures that changed regions remain well-represented and discriminative in the feature space. In our implementation, the reconstruction targets are aligned with the same preprocessing space used by the input images to avoid scale mismatch, and the constraints are activated only when the reconstruction branch is enabled.

D. Memory Bank

Temporal-consistency constraints are effective at suppressing pseudo-changes, but they may also induce an overregularization effect on unchanged regions. In practice, unchanged areas are not visually homogeneous across scenes. For example, the same semantic category such as forest or bare land may exhibit substantially different texture, illumination, and background context under different geographic conditions. When the model is strongly driven to enforce invariance, the representations of unchanged regions tend to collapse toward dominant patterns observed in the training set, which harms cross-scene generalization.

To alleviate this issue, we introduce a cross-sample hierarchical memory bank that stores historical features extracted from diverse training samples and provides retrieval-based references as a soft regularizer. The key idea is to maintain a long-range, cross-scene repository of feature prototypes implicitly learned from data, so that the model can preserve diversity while still being guided toward stable representations. The proposed memory bank acts as a training-time cross-sample regularizer. It stores historical sample-level representations and retrieves relevant feature anchors to guide the current representation. The memory mechanism is removed during inference and therefore does not introduce additional test-time memory operations. The overall cross-sample memory bank and memory-guided alignment process is illustrated in Fig. 3.

1) Hierarchical Memory Design: The memory bank consists of a short-term memory and a long-term memory. The short-term memory captures recently observed feature patterns and adapts quickly to the current training dynamics, while the long-term memory preserves stable and diverse feature anchors accumulated across the whole training set. Both memories are implemented as fixed-size first-in-first-out (FIFO) circular queues and are queried by temperature-scaled cosine

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TABLE I
DEFAULT HYPERPARAMETER CONFIGURATION OF THE HIERARCHICAL MEMORY BANK

Item	Setting
Memory feature stages S	$\{2, 3\}$
Memory embedding dimension D	128
Total memory capacity K	4096
Short-term memory capacity K_s	1024
Long-term memory capacity K_l	3072
Short-/long-term capacity ratio	1:3
Memory projection head ϕ	1×1 Conv + BN + ReLU + GAP
Retrieval number N	10
Similarity temperature τ	0.07
Novelty threshold δ	0.5 ($s_{\max} < \delta$)
Memory-loss weight λ_{mem}	0.2
Short-term update rule	FIFO circular queue
Long-term update rule	Novelty-aware circular queue
Inference behavior	Memory bank disabled

similarity. Their capacities and other default hyperparameters are summarized in Table I. Memory retrieval is performed by temperature-scaled cosine similarity with $\tau = 0.07$. The hierarchical structure provides a natural balance between adaptability and stability, which is crucial for cross-scene generalization.

2) Memory Feature Construction: We construct a compact memory vector from multiscale VMamba features. In our implementation, the selected memory stages are $S = \{2, 3\}$, corresponding to the high-level VMamba stages with stronger semantic representations. Let $\mathbf{X}_i^{t_1} \in \mathbb{R}^{C_i\times H_i\times W_i}$ and $\mathbf{X}_i^{t_2} \in \mathbb{R}^{C_i\times H_i\times W_i}$ denote the stage-wise VMamba features for the two temporal inputs. For each selected stage $i \in \mathcal{S}$, we first compute a temporally stabilized feature by averaging

$$\mathbf{H}_i = \frac{1}{2}\left(\mathbf{X}_i^{t_1} + \mathbf{X}_i^{t_2}\right). \tag{8}$$

This operation suppresses time-specific nuisance variations and yields a more stable descriptor for memory retrieval. We then project H_i into the memory embedding space with a stage-specific projection head $\phi_i(\cdot)$

$$\mathbf{z}_i = \text{GAP}\left(\phi_i\left(\mathbf{H}_i\right)\right) \in \mathbb{R}^D. \tag{9}$$

The memory embedding dimension is set to $D = 128$ in all experiments. Finally, the sample-level memory representation is obtained by averaging the pooled vectors across selected stages

$$\mathbf{z} = \frac{1}{|\mathcal{S}|}\sum_{i\in\mathcal{S}} \mathbf{z}_i. \tag{10}$$

This formulation exactly matches the implementation that stacks pooled vectors from multiple stages and averages them to form $\mathbf{z}$.

3) Retrieval and Reference Aggregation: Given a query z, we compute temperature-scaled cosine similarities to memory entries and retrieve $k = 10$ candidates in total, including 5 candidates from the short-term memory and 5 candidates from the long-term memory

$$s_j = \frac{\mathbf{z}^\top \mathbf{m}_j}{\tau}, \quad w_j = \frac{\exp\left(s_j\right)}{\sum_{l\in\mathcal{N}}\exp\left(s_l\right)}. \tag{11}$$

A reference feature is then formed by a weighted sum

$$\bar{\mathbf{z}} = \sum_{j\in\mathcal{N}} w_j \mathbf{m}_j. \tag{12}$$

Retrieval is performed separately from the short-term and long-term memories. The two retrieved references are fused with weights of 0.6 and 0.4, respectively, to obtain the final historical reference $\bar{z}$. This design provides a stable yet diverse target distribution for regularization, reducing the risk of feature oversmoothing and improving cross-scene transferability.

4) Update Policy and Inference Behavior: During training, the short-term memory is updated with incoming sample features at each iteration using a circular queue. The long-term memory is updated selectively according to a novelty criterion. Given the current memory feature z, we compute its maximum similarity with existing long-term memory entries

$$s_{\max} = \max_{m_j\in\mathcal{M}_l} z^\top m_j. \tag{13}$$

The novelty score is defined as

$$\mathcal{N}(z) = 1 - s_{\max}. \tag{14}$$

If $s_{\max} < 0.5$, equivalently $\mathcal{N}(z) > 0.5$, the current feature is regarded as sufficiently novel and inserted into the long-term memory; otherwise, it only updates the short-term memory.

To mitigate the cold-start problem in the early training stage, we adopt a warm-up strategy for the memory branch. Specifically, the memory-guided loss is disabled during the initial training epochs, and the network is first optimized by the CD loss, deep supervision, and reconstruction constraints. After the feature representations become relatively stable, the memory-guided regularization is activated. Moreover, the memory bank serves only as a soft regularizer, and the retrieval operation is treated as nondifferentiable, so gradients are back-propagated only through the current feature rather than through the stored memory entries. This design reduces the risk of noisy early-stage features misleading the optimization.

The retrieved reference $\bar{\mathbf{z}}$ is explicitly returned by the retrieval procedure and used to build a dedicated memory loss. This loss encourages the current sample representation to align with stable historical references, acting as a soft constraint that complements reconstruction-based invariance without collapsing cross-scene diversity. The complete hyperparameter configuration and algorithmic workflow are provided in Table I and Algorithm 1.

E. Loss Function

C-MNet is trained with a multiterm objective that combines CD supervision, deep supervision, reconstruction-based constraints, and memory-guided regularization

$$\mathcal{L} = \mathcal{L}_{\text{main}} + \mathcal{L}_{\text{ds}} + \lambda_{\text{rec}}\mathcal{L}_{\text{rec}} + \lambda_{\text{ct}}\mathcal{L}_{\text{ct}} + \lambda_{\text{mem}}\mathcal{L}_{\text{mem}}. \tag{15}$$

The main supervision $\mathcal{L}_{\text{main}}$ is applied to the final predicted change map and is instantiated as a weighted combination of binary cross-entropy loss and Dice loss

$$\mathcal{L}_{\text{main}} = 0.7\,\mathcal{L}_{\text{BCE}}\left(\hat{\mathbf{Y}}, \mathbf{Y}\right) + 0.3\,\mathcal{L}_{\text{Dice}}\left(\hat{\mathbf{Y}}, \mathbf{Y}\right). \tag{16}$$

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Algorithm 1 Hierarchical Memory Retrieval and Novelty-Aware Update

Require: Bi-temporal VMamba features $\{X_i^{t1}, X_i^{t2}\}_{i=0}^{3}$; selected stages $\mathcal{S} = \{2, 3\}$; short-term memory $\mathcal{M}_s$; long-term memory $\mathcal{M}_l$; retrieval number $N = 10$; temperature $\tau = 0.07$; novelty threshold $\delta = 0.5$

Ensure: Sample-level representation z and historical reference $\bar{z}$

```
Initialize the candidate list Q ← ∅
for each stage i ∈ S do
    H_i ← 0.5(X_i^{t1} + X_i^{t2})
    z_i ← GAP(φ_i(H_i))
    Add z_i to Q
end for
z ← (1/|S|) Σ_{z_i ∈ Q} z_i
z ← Normalize(z)
Retrieve N/2 nearest entries from M_s using z
Retrieve N/2 nearest entries from M_l using z
Compute similarity weights using temperature τ
Aggregate retrieved entries to obtain z̄_s and z̄_l
z̄ ← 0.6 z̄_s + 0.4 z̄_l
if training mode then
    Insert z into M_s using a FIFO circular queue
    s_max ← max_{m_j ∈ M_l} z^⊤ m_j
    if s_max < δ then
        Insert z into M_l using a FIFO circular queue
    end if
end if
return z and z̄
```

The weights of 0.7 and 0.3 were empirically selected based on validation performance, where BCE provides stable pixel-wise supervision and Dice loss complements it by alleviating class imbalance and improving region-level overlap.

The binary cross-entropy loss for CD is defined as

$$\mathcal{L}_{\mathrm{BCE}}\left(\hat{\mathbf{Y}}, \mathbf{Y}\right) = -\sum_{h,w}\Big[\mathbf{Y}_{h,w}\log\hat{\mathbf{Y}}_{1,h,w} + \left(1-\mathbf{Y}_{h,w}\right)\log\hat{\mathbf{Y}}_{0,h,w}\Big] \tag{17}$$

where $\hat{\mathbf{Y}}_{1,h,w}$ and $\hat{\mathbf{Y}}_{0,h,w}$ denote the predicted probabilities of the changed and unchanged classes at pixel (h, w), respectively, and $\mathbf{Y}_{h,w} \in \{0, 1\}$ is the ground-truth label.

For binary CD, the Dice loss is defined as

$$\mathcal{L}_{\mathrm{Dice}}\left(\hat{\mathbf{Y}}, \mathbf{Y}\right) = 1 - \frac{2\sum_{h,w}\hat{\mathbf{Y}}_{1,h,w}\mathbf{Y}_{h,w} + \epsilon}{\sum_{h,w}\hat{\mathbf{Y}}_{1,h,w} + \sum_{h,w}\mathbf{Y}_{h,w} + \epsilon} \tag{18}$$

where ϵ is a small constant for numerical stability.

The deep supervision term $\mathcal{L}_{\mathrm{ds}}$ denotes the sum of auxiliary losses applied to stage-wise change predictions at different decoder levels. The reconstruction-based constraints $\mathcal{L}_{\mathrm{rec}}$ and $\mathcal{L}_{\mathrm{ct}}$ correspond to the mask-guided dual-temporal reconstruction loss and the cross-temporal consistency loss, respectively. Unless otherwise specified, we set $\lambda_{\mathrm{rec}} = 0.2$, $\lambda_{\mathrm{ct}} = 0.1$, and $\lambda_{\mathrm{mem}} = 0.2$ in all experiments.

For memory-guided regularization, a reference feature $\bar{\mathbf{z}}_b$ is retrieved from the cross-sample hierarchical memory bank for each sample, and its cosine similarity with the current feature $\mathbf{z}_b$ is maximized

$$\mathcal{L}_{\mathrm{mem}} = 1 - \frac{1}{B}\sum_{b=1}^{B}\cos\left(\mathbf{z}_b, \bar{\mathbf{z}}_b\right) \tag{19}$$

where B is the batch size.

For training stability, the memory retrieval operation is treated as nondifferentiable, and gradients are back-propagated only through the current feature $\mathbf{z}_b$. The complete memory construction, retrieval, and novelty-aware update workflow is presented in Algorithm 1.

F. Memory Bank Implementation Details

To provide an actionable and reproducible implementation of the proposed hierarchical memory bank, Table I summarizes its default hyperparameter configuration. Unless otherwise specified, the same settings are used for all datasets. The memory bank has a total capacity of $K = 4096$, including a short-term capacity of $K_s = 1024$ and a long-term capacity of $K_l = 3072$. The complete construction, retrieval, and novelty-aware update process is provided in Algorithm 1.

Algorithm 1 shows that every incoming sample updates the short-term memory, whereas only features satisfying $s_{\max} < \delta$ are consolidated into the long-term memory. Consequently, recent training statistics are continuously retained while redundant long-term entries are suppressed. Memory retrieval and updating are performed only during training. During inference, the memory bank is disabled and therefore introduces no additional memory operations or test-time overhead.

IV. Experiments

A. Data Description

In this experiment, three remote sensing image CD datasets commonly used in CD tasks were adopted, namely the LEVIR-CD dataset [2], WHU-CD [44], and the SYSU-CD [35], to conduct comparative and ablation experiments. These datasets contain diverse land-cover types, complex scene distributions, multiscale change patterns, and substantial intraclass appearance variations in unchanged regions.

1) SYSU-CD Dataset [35]: The SYSU-CD dataset contains 20 000 pairs of aerial images collected in Hong Kong from 2007 to 2014. Each image has a size of 256 × 256 pixels with a spatial resolution of 0.5 m. This dataset covers diverse change types, including new urban construction, urban expansion, abandoned construction sites, vegetation changes, road expansion, and offshore construction. Unlike datasets mainly focused on building changes, SYSU-CD contains complex scenes involving changes in various land-cover categories, such as buildings, rivers, water surfaces, and forests. Such diversity and scene complexity make SYSU-CD a challenging benchmark for CD algorithms. In our experiments, the dataset was divided into 12 000 training image pairs, 4000 validation image pairs, and 4000 testing image pairs.

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2) WHU-CD Dataset [44]: The WHU-CD dataset is a building CD dataset constructed from aerial images of Christchurch, New Zealand, after the 2011 earthquake. The core data were acquired in 2012, covering approximately 20.5 km^2 and containing 12 796 buildings. The aerial images have a spatial resolution of 0.075 m and were geometrically corrected using 30 ground control points (GCPs), achieving an accuracy of approximately 1.6 pixels. In our experiments, the original image pairs were cropped into 256 × 256 patches, resulting in 6096 training patches, 762 validation patches, and 762 testing patches.

3) LEVIR-CD Dataset [2]: The LEVIR-CD dataset consists of 673 pairs of high-resolution bi-temporal image patches collected from Google Earth, with a spatial resolution of 0.5 m per pixel. Each image pair has a size of 1024 × 1024 pixels. In our experiments, the original images are nonoverlappingly cropped into 256 × 256 patches, following the standard data split protocol, with 7120 samples for training, 1024 for validation, and 2048 for testing.

The temporal interval between image pairs ranges from 5 to 14 years, during which significant land-use changes and intensive urban construction activities have occurred. Geographically, the dataset covers 20 different regions in Texas, USA, including cities such as Austin and Buda, providing diverse urban scenes and construction patterns.

Despite its wide applicability, LEVIR-CD poses notable challenges for CD algorithms. In particular, building boundaries are often affected by illumination variations and shadow effects, which weaken edge feature representations and may lead to FPs and false negatives (FNs) in boundary regions.

B. Experimental Setup

1) Implementation Details: All experiments were conducted on a workstation equipped with an NVIDIA RTX 3090 GPU. The proposed method was implemented using the PyTorch 2.1.1 deep learning framework, with CUDA 11.8 for GPU acceleration. The network was optimized using the AdamW optimizer, with an initial learning rate of 1×10^{-4} to balance convergence speed and training stability. The weight decay coefficient was set to 5×10^{-4}, and the model was trained for 50 epochs. Model checkpoints were saved at each epoch to facilitate model selection and prevent overfitting. The batch size was set to 8. A polynomial learning rate decay strategy was applied during training. Standard data augmentation strategies, including random horizontal and vertical flipping, were applied during training.

2) Evaluation Metrics: In this experiment, we used multiple evaluation metrics to assess the model's performance. While Recall and Precision are standard metrics, they can be somewhat misleading in cases of unbalanced classes. Therefore, we primarily refer to intersection over union (IoU), $F1$ score, and Kappa coefficient (KC) for a more robust evaluation. Here, TP represents true positives, FN represents false negatives, FP represents false positives, and TN represents true negatives.

a) Recall rate: Recall measures the model's ability to find actual changes. The formula is

$$\text{Recall} = \frac{TP}{TP + FN}. \tag{20}$$

b) Precision rate: Precision indicates the proportion of correct changes among predicted changes. The formula is

$$\text{Precision} = \frac{TP}{TP + FP}. \tag{21}$$

c) F1 score: $F1$-score is the harmonic mean of precision and recall. The formula is

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}. \tag{22}$$

d) Intersection over union: IoU measures the overlap of predicted and actual changes. The formula is

$$IoU = \frac{TP}{TP + FP + FN}. \tag{23}$$

e) Kappa coefficient: Kappa measures classification consistency. The formula is

$$\text{Kappa} = \frac{p_o - p_e}{1 - p_e} \tag{24}$$

where p_o is the actual classification accuracy, and p_e is the expected accuracy of random classification.

C. Comparison With State-of-the-Art Methods

1) FC-Siam-Diff [33]: FC-Siam-diff adopts a Siamese architecture with two identical branches to process bi-temporal images independently. During decoding, the absolute differences between the corresponding skip connections of the two branches are computed and concatenated to highlight change-related information.

2) FC-Siam-Conc [33]: FC-Siam-conc also employs a weight-sharing Siamese network to extract features from two temporal images. Instead of differencing, the skip connection features from both branches are directly concatenated in the decoding stage to generate change predictions.

3) SRCDNet [34]: SRCDNet is designed for CD between images with different spatial resolutions. It first enhances low-resolution images using a super-resolution module and then adopts a Siamese architecture with attention mechanisms to extract features and perform CD.

4) DSAMNet [45]: DSAMNet utilizes a Siamese network to extract multiscale features and enhances feature discriminability through metric learning and the convolutional block attention module (CBAM). In addition, deep supervision and a hybrid loss function are employed to improve optimization and detection accuracy.

5) BIT [36]: BIT represents bi-temporal images as a compact set of visual tokens and employs a Transformer encoder to model contextual relationships in the token space. The enriched tokens are projected back to the pixel space through a Transformer decoder, and a shallow CNN is applied to the feature difference maps to produce pixel-level change predictions.

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6) MSCANet [46]: MSCANet combines CNNs and Transformers for CD. CNN-based modules extract hierarchical features, while a multiscale context aggregator (MSCA) within the Transformer framework captures global contextual information. A multibranch prediction head further enhances the learning of deep-layer features.

7) DDCNN [12]: DDCNN consists of convolutional units, upsampling attention (UA) units, and difference enhancement (DE) units. The convolutional units extract features, UA units enhance texture and detail information during upsampling, and DE units explicitly process difference maps between bitemporal images.

8) DMINet [13]: DMINet is built upon two parallel Siamese pretrained CNNs, where dual-temporal features interact through a temporal joint attention module. Pixel-level subtraction and channel-level concatenation are jointly employed to generate difference features, which are progressively aligned across scales using Hadamard product operations.

9) AANet [47]: AANet adopts ResNet-18 as the backbone and addresses pseudo-change ambiguities through an ambiguity refinement module (ARM). ARM consists of an ambiguity and difference extractor (ADE) and an ambiguity-guided difference refiner (AGDR), which collaboratively extract ambiguous regions and refine difference features.

10) HRMNet [48]: HRMNet is a deep learning framework that improves CD performance in hard regions, such as change boundaries, small objects, and pseudo-changes. It introduces a cross-layer knowledge review module and an online hard-region estimation branch to enhance feature learning in challenging areas.

D. Comparing Experiments

In this section, we comprehensively evaluated the performance of multiple methods introduced in Section IV-C, including our proposed approach, on three different datasets, using a combination of quantitative and qualitative analyses. Their results are included in the quantitative tables to provide a complete numerical overview of all compared techniques.

Quantitative results are presented in tables, with relevant metrics already described in Section IV-B2. Qualitative results are shown directly via images. In these images, white denotes true positive (TP, correctly identifying positive samples as positive), red represents FP (misclassifying negative samples as positive), black indicates true negative (TN, correctly identifying negative samples as negative), and cyan stands for FN (misclassifying positive samples as negative). Each dataset is associated with one table and one image, and detailed descriptions will be given for each dataset hereinafter.

1) Comparative Experiments on SYSU-CD Dataset: Table II reports the quantitative comparison results of different CD methods on the SYSU-CD dataset. Five widely used evaluation metrics are adopted, including Recall (Rec), Precision (Pre), $F1$-score ($F1$), IoU, and KC. For each metric, the top two results are highlighted in red and green, respectively.

From Table II, it can be observed that traditional Siamese CNN-based methods, such as FC-Siam-diff and FC-Siam-conc, achieve relatively high precision but suffer from limited recall, indicating their sensitivity to pseudo-changes and weak generalization under complex scene variations. Methods incorporating attention mechanisms or multiscale modeling, including DSAMNet, MSCANet, and DMINet, show consistent improvements across most metrics, demonstrating the effectiveness of enhancing feature discrimination and contextual aggregation.

Transformer-based or hybrid architectures, such as BIT and HRMNet, further improve performance by modeling long-range dependencies and cross-temporal interactions. In particular, HRMNet achieves strong results in terms of

TABLE II
COMPARATIVE EXPERIMENT RESULTS FOR EACH COMPARISON METHOD ON SYSU-CD DATASET. THE TOP TWO RESULTS ARE HIGHLIGHTED IN RED AND GREEN

Methods	Rec(%)	Pre(%)	F1(%)	IoU(%)	KC(%)
FC-Siam-diff	58.18	90.53	70.84	54.84	64.24
FC-Siam-conc	72.92	80.44	76.49	61.94	69.70
SRCDNet	78.60	73.57	76.00	61.29	68.27
DSAMNet	82.16	74.88	78.35	64.41	71.26
BIT	76.19	81.90	78.94	65.21	72.74
MSCANet	75.70	81.33	78.42	64.49	72.06
DDCNN	78.88	81.82	80.32	67.12	74.40
DMINet	79.76	84.13	81.89	69.33	76.49
AANet	79.81	84.55	82.11	69.66	77.46
HRMNet	80.81	83.97	82.36	70.01	77.05
Ours	81.68	86.71	84.12	72.59	78.28

TABLE III
COMPARATIVE EXPERIMENT RESULTS FOR EACH COMPARISON METHOD ON WHU-CD DATASET. THE TOP TWO RESULTS ARE HIGHLIGHTED IN RED AND GREEN

Methods	Rec(%)	Pre(%)	F1(%)	IoU(%)	KC(%)
FC-Siam-diff	91.16	80.76	85.65	74.89	84.97
FC-Siam-conc	92.60	75.40	83.12	71.12	82.29
SRCDNet	91.90	89.51	90.69	82.96	90.27
DSAMNet	78.94	81.58	80.23	66.99	79.37
BIT	92.92	93.59	93.26	87.37	92.96
MSCANet	91.99	88.20	90.05	81.91	89.60
DDCNN	87.93	96.34	91.95	85.09	91.60
DMINet	89.19	87.02	88.09	78.72	87.55
AANet	89.84	79.69	83.64	73.18	82.93
HRMNet	87.15	92.70	89.84	81.55	89.40
Ours	95.25	93.20	94.21	89.06	93.84

TABLE IV
COMPARATIVE EXPERIMENT RESULTS FOR EACH COMPARISON METHOD ON LEVIR-CD DATASET. THE TOP TWO RESULTS ARE HIGHLIGHTED IN RED AND GREEN

Methods	Rec(%)	Pre(%)	F1(%)	IoU(%)	KC(%)
FC-Siam-diff	83.16	90.05	86.47	76.16	85.77
FC-Siam-conc	84.66	89.14	86.84	76.75	86.16
SRCDNet	90.71	81.49	85.85	75.21	85.05
DSAMNet	89.09	80.23	84.43	73.05	83.54
BIT	86.98	92.16	89.50	80.99	88.95
MSCANet	89.10	90.41	89.75	81.40	89.20
DDCNN	89.08	92.18	90.60	82.82	90.10
DMINet	89.00	92.64	90.78	83.12	90.30
AANet	90.25	91.29	90.77	83.13	88.30
HRMNet	88.69	90.53	89.92	81.69	89.39
Ours	90.88	92.34	91.60	84.51	90.79

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TABLE V

ABLATION STUDY OF RECONSTRUCTION AND MEMORY MODULES FOR ENHANCING UNCHANGED CONSISTENCY ON SYSU-CD, WHU-CD, AND LEVIR-CD DATASETS

RECON	MEMORY	SYSU			WHU			LEVIR		
		F1(%)	IoU(%)	KC(%)	F1(%)	IoU(%)	KC(%)	F1(%)	IoU(%)	KC(%)
base		82.68	70.48	77.23	92.27	85.66	91.93	90.65	82.89	90.16
✓		83.12	71.12	77.35	93.11	87.10	92.80	90.80	83.14	90.32
	✓	83.76	72.06	77.66	92.81	86.59	92.50	90.96	83.42	90.49
✓	✓	84.12	72.59	78.28	94.21	89.06	93.84	91.60	84.51	90.79

TABLE VI

COMPARISON OF MODEL COMPLEXITY AND INFERENCE EFFICIENCY ON THE SYSU-CD DATASET. FLOPS ARE CALCULATED FOR ONE PAIR OF 256 × 256 BITEMPORAL IMAGES, WHILE INFERENCE TIME IS MEASURED ON THE FULL TEST SET UNDER THE SAME HARDWARE AND SOFTWARE ENVIRONMENT

Method	**Params (M)**	**FLOPs (G)**	**Inference Time (s)**	**F1 (%)**	**IoU (%)**
FC-Siam-diff	1.35	4.72	23.0	70.84	54.84
FC-Siam-conc	1.55	5.32	23.0	76.49	61.94
SRCDNet	16.98	71.68	43.0	76.00	61.29
DSAMNet	16.95	75.39	62.5	78.35	64.41
BIT	11.47	26.31	25.4	78.94	65.21
MSCANet	16.42	14.80	31.0	78.42	64.49
DDCNN	46.68	177.44	148.0	80.32	67.12
DMINet	6.24	14.55	57.0	81.89	69.33
AANet	15.82	24.21	45.0	82.11	69.66
HRMNet	13.44	11.54	27.0	82.36	70.01
C-MNet	76.60	52.49	76.0	**84.12**	**72.59**

TABLE VII

ALL-PAIRS SINGLE-SOURCE CROSS-SCENE GENERALIZATION RESULTS ON SYSU-CD. EACH MODEL IS TRAINED ON ONE SOURCE SCENE AND DIRECTLY EVALUATED ON THE OTHER THREE UNSEEN SCENES WITHOUT FINE-TUNING

Source	Target	HRMNet		Ours	
		IoU	F1	IoU	F1
Dense	Suburban	39.19	56.31	72.83	84.28
	Vegetation	54.42	70.48	58.39	73.73
	Trans./Coastal	72.83	84.28	80.67	89.31
Suburban	Dense	58.36	73.70	65.80	79.37
	Vegetation	52.82	69.13	58.69	73.96
	Trans./Coastal	74.51	85.39	87.56	93.37
Vegetation	Dense	67.92	80.90	72.53	84.07
	Suburban	44.18	61.29	68.07	81.00
	Trans./Coastal	59.19	74.37	67.98	80.94
Trans./Coastal	Dense	62.68	77.06	70.88	82.96
	Suburban	41.22	58.37	82.81	90.59
	Vegetation	52.97	69.25	56.52	72.22
Macro Average		56.69	71.71	**70.23**	**82.15**

TABLE VIII

ABLATION STUDY ON KEY ARCHITECTURAL DESIGNS ON SYSU-CD

Method	Visual prior	Temporal interaction	F1	IoU
VMamba only	None	Proposed	82.87	70.74
ResNet prior	ResNet	Proposed	82.92	70.83
Late subtraction	Frozen SAM2	Late subtraction	83.76	72.06
Ours	Frozen SAM2	Proposed	**84.12**	**72.59**

$F1$-score and IoU, benefiting from its cross-layer knowledge review strategy for hard regions.

Overall, the proposed method achieves the best performance on SYSU-CD, obtaining the highest $F1$-score (84.12%), IoU (72.59%), and KC (78.28%), while maintaining competitive precision. Compared with the second-best method, our approach improves IoU by a noticeable margin, indicating more accurate localization of changed regions under complex and diverse scene conditions.

Fig. 4 presents qualitative comparisons on several representative samples from the SYSU-CD dataset. The selected scenes include diverse land-cover types such as water bodies, vegetation, urban areas, and their complex boundaries, which are challenging for CD due to background clutter and subtle appearance variations.

As shown in Fig. 4, early Siamese CNN-based methods (e.g., FC-Siam-diff and FC-Siam-conc) tend to produce fragmented predictions and suffer from noticeable false positives along object boundaries and background regions. Methods incorporating attention mechanisms or multiscale feature modeling reduce some false alarms but still miss parts of changed regions, resulting in evident FNs, especially in narrow or irregular structures.

Transformer-based and hybrid methods (e.g., BIT, HRMNet) improve the overall structural consistency; however, they still exhibit either overdetection in homogeneous areas or

TABLE IX

COMPARISON OF FROZEN AND FINE-TUNED SAM2 ON SYSU-CD. F AND FT DENOTE FROZEN AND FINE-TUNED, RESPECTIVELY. MEM. DENOTES TOTAL PEAK TRAINING MEMORY ACROSS TWO GPUS, AND TIME DENOTES TRAINING TIME PER EPOCH

Setting	F1	IoU	Mem. (GB)	Time (s/ep.)
SAM2-S (F)	81.29	68.47	7.91	205.3
SAM2-S (FT)	82.66	70.45	41.86	538.8
SAM2-L (F)	**84.12**	**72.59**	7.96	601.4

TABLE X

SENSITIVITY ANALYSIS OF THE MEMORY BANK ON SYSU-CD

Setting	Value	F1	IoU
Memory capacity K	2048	83.11	71.10
	4096	**84.12**	**72.59**
	8192	83.03	70.98
Short-/long-term ratio	1:1	82.35	70.00
	1:3	**84.12**	**72.59**
	3:1	83.28	71.35
Retrieval number N	5	83.42	71.56
	10	**84.12**	**72.59**
	20	82.59	70.34
λ_{mem}	0.10	83.09	71.07
	0.20	**84.12**	**72.59**
	0.30	83.39	71.51

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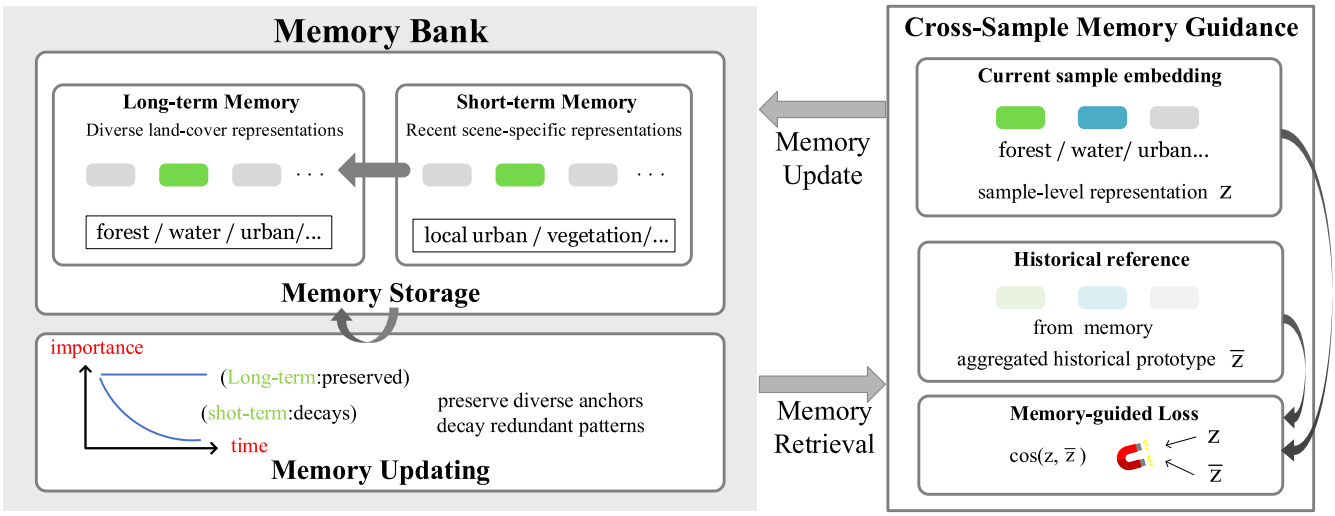


Fig. 3. Cross-sample memory bankand memory-guided alignment. The memory bank maintains both long-term and short-term representations of unchanged regions across different scenes, preserving diverse land-cover patterns while suppressing redundant or scene-specific variations. Given a current sample embedding, representative features are retrieved from the memory bank to form an aggregated historical reference, which guides the alignment of current representations through a memory-guided loss. The memory bank is used only during training to provide cross-sample historical references for memory-guided regularization and introduces no additional memory operations during inference.

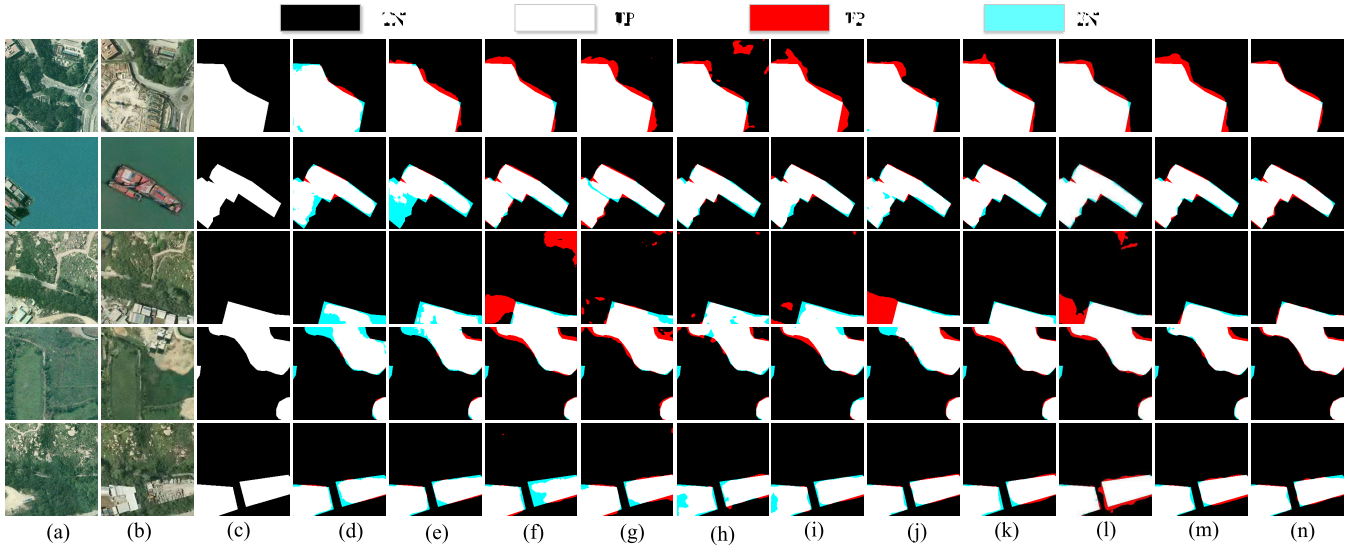


Fig. 4. Qualitative results of the overall methods on the SYSU-CD dataset. The white, black, red, and cyan colors represent TP, TN, FP, and FN results, respectively. (a) Image T1. (b) Image T2. (c) Ground truth. (d) FC-Siam-diff. (e) FC-Siam-conc. (f) SRCDNet. (g) DSAMNet. (h) BIT. (i) MSCANet. (j) DDCNN. (k) DMINet. (l) AANet. (m) HRMNet. (n) Ours.

under-detection in complex scenes. In contrast, the proposed method generates more complete and coherent change regions while effectively suppressing false positives. The boundaries of changed objects are better preserved, and fewer FNs are observed in challenging areas.

These qualitative results are consistent with the quantitative improvements reported in Table II, demonstrating the effectiveness of the proposed method in balancing accurate change localization and robustness under complex scene variations.

2) Comparative Experiments on WHU-CD Dataset: We list the various metrics of multiple comparative methods on the WHU-CD dataset in Table III. WHU-CD is particularly challenging due to its large-scale scenes, significant variation in building sizes, and complex background structures. Traditional Siamese-based methods, such as FC-Siam-diff and FC-Siam-conc, achieve relatively low precision and IoU scores, indicating their limited robustness to pseudo-changes and boundary ambiguities in large-area aerial images.

Methods incorporating attention mechanisms or global context modeling (e.g., DSAMNet, BIT, and MSCANet) show noticeable improvements, especially in recall and $F1$-score, demonstrating their enhanced ability to capture long-range dependencies. However, these approaches still struggle to maintain a balanced trade-off between false positives and missed detections, particularly in regions with complex land-cover transitions.

The proposed method achieves the best overall performance on WHU-CD, obtaining the highest Recall, $F1$-score, IoU,

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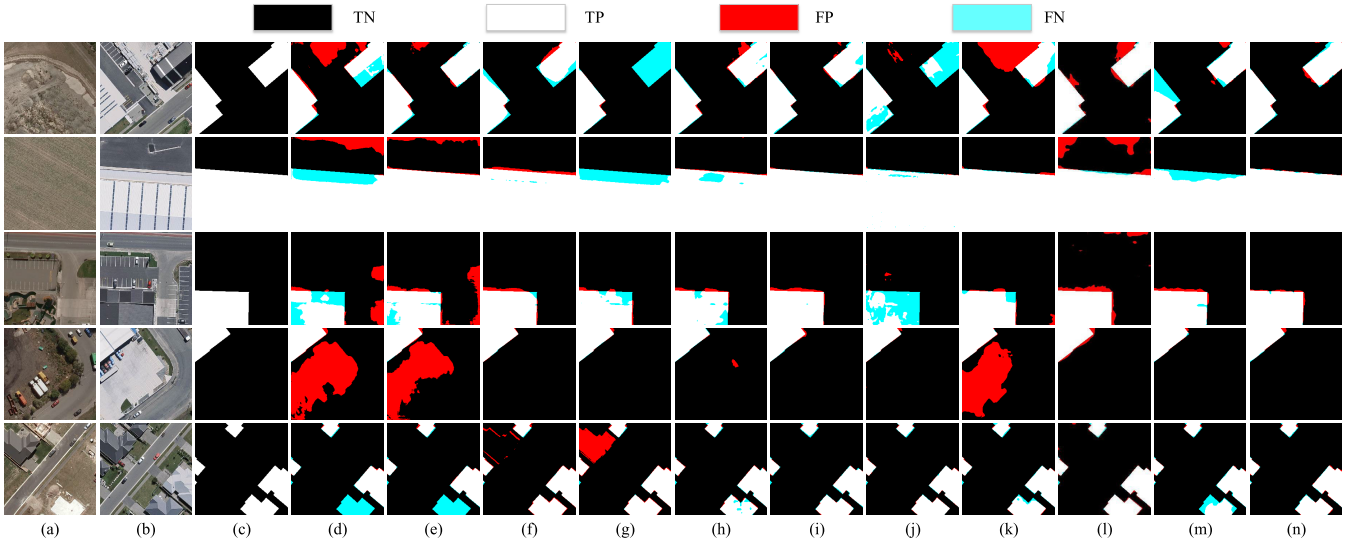


Fig. 5. Qualitative results of the overall methods on the WHU-CD dataset. The white, black, red, and cyan colors represent TP, TN, FP, and FN results, respectively. (a) Image T1. (b) Image T2. (c) Ground truth. (d) FC-Siam-diff. (e) FC-Siam-conc. (f) SRCDNet. (g) DSAMNet. (h) BIT. (i) MSCANet. (j) DDCNN. (k) DMINet. (l) AANet. (m) HRMNet. (n) Ours.

and KC, while maintaining competitive precision. Notably, it obtains the highest IoU and KC values, indicating more accurate spatial localization and stronger agreement with ground truth. These results demonstrate that the proposed combination of temporal consistency constraints and memory-guided feature regularization effectively enhances both detection accuracy and generalization capability on large-scale and complex CD scenarios.

We present the qualitative results of multiple comparative methods on the WHU-CD dataset in Fig. 5. Each row represents a set of comparison images. As shown in the first two columns, WHU-CD contains large-scale building changes with complex backgrounds and varying spatial structures, which make accurate boundary delineation particularly challenging.

Siamese-based methods, such as FC-Siam-diff and FC-Siam-conc, tend to generate fragmented predictions and obvious false alarms near building boundaries and shadow regions. Although attention-based and context-aware approaches (e.g., DSAMNet, BIT, and MSCANet) alleviate part of these issues, they still exhibit noticeable false positives in cluttered regions and missed detections in small or irregular structures.

In contrast, the proposed method produces more compact and complete change regions with clearer boundaries and fewer spurious responses. It effectively suppresses pseudo-changes in homogeneous areas while preserving fine structural details in complex scenes. Benefiting from the reconstruction-based temporal consistency constraint and memory-guided regularization, our approach demonstrates superior robustness to background variations and scale changes, leading to more reliable change maps on challenging WHU-CD scenes.

3) Comparative Experiments on LEVIR-CD Dataset: We present the performance metrics of each comparative method on the LEVIR-CD dataset in Table IV. LEVIR-CD mainly focuses on building CD in high-resolution aerial images, where accurate boundary localization and structural completeness are critical.

Traditional Siamese CNN-based methods, such as FC-Siam-diff and FC-Siam-conc, achieve reasonable recall but exhibit limited precision, indicating a tendency to introduce false positives along building edges. Methods incorporating attention mechanisms or multiscale modeling, including SRCDNet and DSAMNet, improve feature discrimination but still struggle to balance recall and precision.

Transformer-based and hybrid architectures, such as BIT, MSCANet, DDCNN, and DMINet, further enhance global context modeling and yield more balanced performance across metrics. Among these methods, DMINet achieves competitive $F1$ and KC scores, demonstrating the effectiveness of temporal interaction mechanisms.

The proposed method achieves the best overall performance on LEVIR-CD, obtaining the highest $F1$-score (91.60%), IoU (84.51%), and KC (90.79%). Compared with other methods, it achieves a favorable balance between recall and precision, resulting in more accurate building change localization with fewer false alarms. These results demonstrate that the proposed framework effectively captures building-specific change patterns while preserving boundary integrity in high-resolution scenes.

We present the qualitative results of multiple comparative methods on the LEVIR-CD dataset in Fig. 6. Each row represents a set of comparison images. As shown in Fig. 6, different methods exhibit distinct error patterns on building CD. Siamese CNN-based approaches, such as FC-Siam-diff and FC-Siam-conc, tend to produce fragmented change regions and false positives along building boundaries, mainly due to their limited ability to model complex structural variations. Methods incorporating attention mechanisms or global context modeling (e.g., SRCDNet, DSAMNet, and BIT) alleviate some boundary ambiguities, but still suffer from missed detections in small or irregularly shaped buildings.

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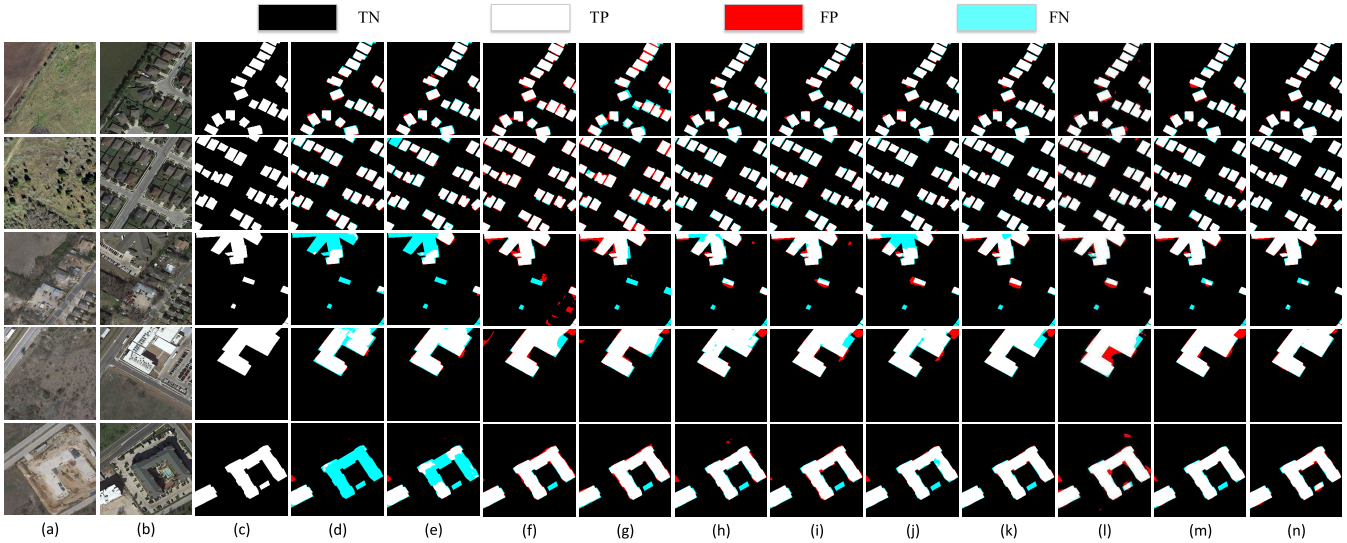


Fig. 6. Qualitative results of the overall methods on the LEVIR-CD dataset. The white, black, red, and cyan colors represent TP, TN, FP, and FN results, respectively. (a) Image T1. (b) Image T2. (c) Ground truth. (d) FC-Siam-diff. (e) FC-Siam-conc. (f) SRCDNet. (g) DSAMNet. (h) BIT. (i) MSCANet. (j) DDCNN. (k) DMINet. (l) AANet. (m) HRMNet. (n) Ours.

Compared with these methods, the proposed approach generates more complete and compact building change regions with fewer false positives. In particular, boundary regions and small-scale building changes are better preserved, and pseudo-changes caused by appearance variations are effectively suppressed. These observations indicate that the proposed method benefits from both temporal consistency constraints and memory-guided feature regularization, leading to more robust and accurate building CD in high-resolution scenes. The component-wise ablation results are summarized in Table V and further discussed in Section IV-E.

4) Model Complexity and Efficiency Analysis: Table VI compares the model complexity and inference efficiency of different methods on the SYSU-CD dataset. C-MNet achieves the highest $F1$ and IoU scores, outperforming the second-best HRMNet by 1.76 and 2.58 percentage points, respectively. Although the dual-branch architecture introduces additional computational overhead, its FLOPs and inference time remain substantially lower than those of DDCNN. Moreover, freezing the SAM2 trunk avoids gradient computation and parameter updates during training. Overall, C-MNet provides a reasonable trade-off between detection accuracy and computational cost.

5) Cross-Scene Generalization: To specifically evaluate the claimed "cross-scene" generalization capability, we further construct a scene-disjoint evaluation protocol on SYSU-CD. Different from the conventional random patch-level split, which mainly evaluates performance under similar scene distributions, we divide the samples into four representative scene types according to their dominant land-cover composition and spatial characteristics: Dense Urban, Suburban/Mixed, Vegetation/Open Land, and Transportation/Coastal. Representative examples of these scene types are shown in Fig. 7, where clear differences in land-cover composition and spatial structure can be observed.

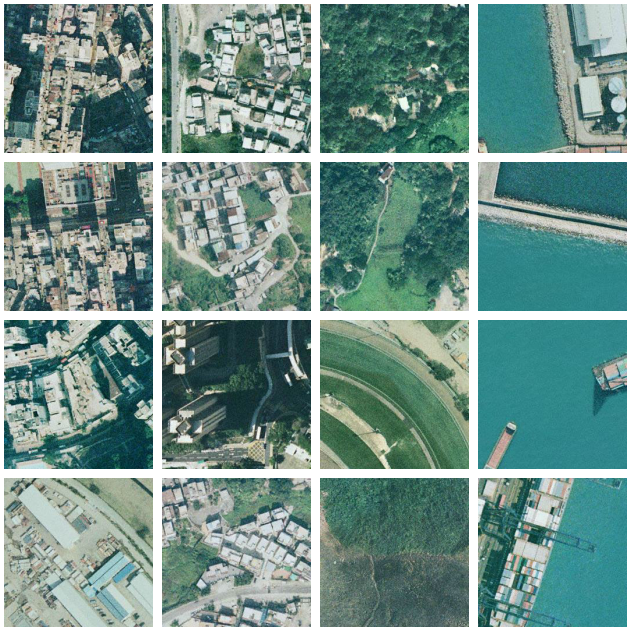


Fig. 7. Representative examples of the four scene types in SYSU-CD. (a) Dense urban. (b) Suburban/mixed. (c) Vegetation/open land. (d) Transportation/coastal.

We adopt an all-pairs single-source cross-scene evaluation protocol. Specifically, each scene type is independently used as the source domain for training and validation, while the remaining three scene types are exclusively used as unseen target domains. Therefore, four models are trained for each method, resulting in 12 source-to-target evaluation settings. No samples from the target scenes are involved in training, validation, model selection, or fine-tuning. HRMNet and our method use identical data partitions and experimental settings.

As shown in Table VII, the proposed method consistently outperforms HRMNet across all twelve source-target scene

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combinations. Specifically, the macro-average IoU increases from 56.69% to 70.23%, while the $F1$-score improves from 71.71% to 82.15%, corresponding to gains of 13.54 and 10.44 percentage points, respectively. Consistent improvements are observed across different transfer directions, including transfers between urban-dominated, vegetation-dominated, and transportation/coastal scenes.

These results demonstrate that the proposed model is less dependent on source-specific scene characteristics and maintains stronger robustness when transferred to previously unseen scene distributions. In particular, the reconstruction constraint promotes stable temporal representations, while the cross-sample memory bank introduces diverse feature patterns beyond the current source samples. Their combination therefore provides a better balance between feature consistency and diversity, contributing to improved cross-scene generalization.

E. Ablation Experiments

1) Effect of Main Components: Table V reports the ablation results of the proposed reconstruction constraint and memory bank on three benchmark datasets. Starting from the baseline model without additional consistency regularization, introducing the reconstruction branch consistently improves the $F1$-score and IoU across all datasets. This demonstrates that mask-guided reconstruction effectively enforces temporal invariance in unchanged regions and suppresses pseudo-changes caused by illumination and seasonal variations.

As shown in Fig. 1, the reconstruction module substantially reduces false positives in complex unchanged regions compared with the baseline. However, the resulting consistency regularization may also oversmooth some genuinely changed regions, leading to missed detections.

The memory bank also improves performance when used independently, indicating that cross-sample historical representations provide useful feature diversity beyond the current mini-batch. When combined with the reconstruction constraint, the model achieves the best performance on all three datasets. This suggests that reconstruction-based temporal consistency and memory-based cross-sample regularization are complementary: the former stabilizes representations against temporal nuisance variations, while the latter preserves diverse historical feature patterns and mitigates overregularization.

2) Ablation on Key Architectural Designs: To evaluate the necessity of the SAM2 branch and the proposed cross-temporal interaction, we construct three variants: removing SAM2 and retaining only VMamba, replacing SAM2 with a pretrained ResNet encoder, and replacing the proposed interaction with late feature subtraction.

As shown in Table VIII, removing the SAM2 branch or replacing it with ResNet reduces the $F1$/IoU scores from 84.12/72.59–82.87/70.74 and 82.92/70.83, respectively, confirming the benefit of the generic visual representations provided by SAM2. Replacing the proposed cross-temporal interaction with late subtraction also decreases the $F1$ and IoU scores to 83.76 and 72.06, demonstrating that explicit interaction before feature scanning is more effective than independent processing followed by feature differencing.

3) Effect of Freezing and Fine-Tuning SAM2: As shown in Table IX, fine-tuning SAM2-Small improves the $F1$ and IoU scores by 1.37 and 1.98 percentage points, respectively. However, it increases the total training memory from 7.91 to 41.86 GB and the training time from 205.3 to 538.8 s per epoch because gradients, intermediate activations, and optimizer states must be maintained for the SAM2 trunk.

In comparison, frozen SAM2-Large achieves the highest accuracy and requires substantially less training memory than fine-tuned SAM2-Small, although its larger backbone results in a longer forward-processing time. Therefore, we adopt frozen SAM2-Large to exploit its stronger transferable representations without back-propagating through the pretrained trunk.

4) Sensitivity Analysis of the Memory Bank: Table X presents a one-factor-at-a-time sensitivity analysis, where one hyperparameter is varied while the others remain fixed. The best performance is obtained with a total memory capacity of $K = 4096$, a capacity ratio of $K_s{:}K_l = 1{:}3$, a retrieval number of $N = 10$, and $\lambda_{\text{mem}} = 0.2$.

A small memory limits the diversity of stored feature anchors, whereas an excessively large memory may introduce redundant entries. Allocating more capacity to long-term memory provides a better balance between recent adaptation and stable feature preservation. Moreover, retrieving too many entries may introduce irrelevant features, while an excessively large λ_{mem} may over constrain feature learning.

V. Conclusion

In this article, we presented C-MNet, a constraint- and memory-driven framework for bi-temporal remote sensing CD. By integrating a frozen SAM2 spatial branch and a VMamba temporal branch, the proposed network effectively captures both spatial structures and cross-temporal dependencies. Furthermore, a mask-guided dual-temporal reconstruction module is introduced to explicitly enforce temporal consistency in unchanged regions, thereby suppressing pseudo-changes caused by illumination, seasonal variations, and sensor discrepancies.

To balance consistency and diversity, we further design a cross-sample hierarchical memory bank that stores stable sample-level representations across different scenes and provides reliable historical references during training. This mechanism prevents feature collapse induced by excessive consistency constraints and enhances the model's cross-scene generalization capability. Extensive experiments on the WHU-CD, SYSU-CD, and LEVIR-CD datasets demonstrate that C-MNet consistently outperforms state-of-the-art methods in terms of accuracy, robustness, and generalization.

Although promising results have been achieved, several directions remain for future research. First, the current memory updating strategy can be further optimized to better adapt to complex scene distributions. Second, extending the proposed framework to heterogeneous and multimodal CD scenarios is a valuable direction. Finally, exploring more lightweight architectures may facilitate the deployment of C-MNet in large-scale and real-time applications.

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